

ENGINEERING

Specifically, you should research several different *products* or *technologies*

- **Environmental impact of technology in the field** (positive or negative impacts, ways to reduce negative impacts, agencies that provide resources etc)

[Engineering's Environmental Impact: Shaping Our Planet. For Better Or Worse | ShunWaste](#)

Positive Impacts:

MORE BROADLY:

[What is BIM? | Building Information Modeling | Levels of BIM | Trimble Resource Center](#)

- Engineers develop tech that improves energy efficiency, reduce waste, and support sustainable infrastructure
- Computer modeling and simulation help reduce material use by identify design flaws before construction
- Smart systems optimize resource use, reducing electricity, water, and fuel consumption
- BIM allows engineers to plan projects more efficiently, reducing construction waste and improving long-term building performance
- Usage of computer-aided design (CAD) and simulation software
 - a) Allows for testing of structures, machines, and systems digitally before building physical prototypes (reduces material waste, lowers energy consumption, and minimizes failed bulbs that would otherwise require reconstruction.)
 - b) In civil and mechanical engineering, simulation software can test stress, airflow, and structural safety without needing multiple physical models
- Development of smart systems and sensors
 - a) Helps balance energy demand and reduce electricity waste, while environmental sensors in civil engineering monitor air quality, water quality, and pollution levels in real time
 - b) Allows for engineers to respond quickly to environmental risks and improve sustainability
- Building Information Modelling (BIM)
 - a) Digital representation of physical and functional characteristics of buildings or other physical assets (construction of a structure or building within one 3D model)
 - b) Allows for governments, municipalities, and property managers to make informed decisions based on information gathered from the mode, even after the building is constructed.

Negative Impacts:

- Pollution from Manufacturing and construction activities
- Habitation destruction and ecosystem disruption during infrastructure development
- Extraction of raw materials such as metals, minerals, and fossil fuels
- High energy consumption from industrial processes and data centers
- Electronic waste generated by outdated technologies

Pollution, ecosystem disruption, natural resources' extraction

Ways to Reduce Negative Impacts

- Use renewable and recyclable materials

- Design products for durability and repairability
- Implement energy-efficiency tech and smart grids
- Use CAD, simulation software, and BIM to reduce material waste and construction errors
- Conduct Environment Impact Assessments (EIAs) before major projects
- Follow sustainable engineering standards such as LEED certification and ISO 14001 environmental management systems

Agencies that Provide Resources:

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Specific Example:

Trend:

- Major tren